

# Embedded Linux & System Software Interview Handbook

## Phase 7 – Linux Security & Secure Boot

This phase introduces Linux kernel security, secure boot mechanisms and platform hardening techniques used in modern embedded devices, servers and cloud systems.

## Chapter 14 – Linux Security & Secure Boot

### *Topics Covered*

- Linux Security Architecture
- Kernel Security Model
- User, Group and Permission Model
- POSIX ACLs
- Linux Capabilities
- Namespaces and Isolation
- cgroups Security
- LSM (Linux Security Modules)
- SELinux
- AppArmor
- Seccomp
- Secure Boot
- UEFI Secure Boot
- Measured Boot
- Trusted Platform Module (TPM)
- Disk Encryption (dm-crypt, LUKS)
- Kernel Lockdown
- IMA (Integrity Measurement Architecture)
- EVM (Extended Verification Module)
- Cryptographic APIs
- Random Number Generation
- Certificate and Key Management
- Container Security
- Kubernetes Security Basics
- Trusted Execution Environments (TEE)
- Confidential Computing Overview
- Vulnerability Mitigation (ASLR, NX, Stack Canaries)
- Kernel Hardening

- Security Debugging and Auditing
- Real Production Case Studies
- Senior Interview Questions

## Learning Outcomes

After completing this phase, you will understand how Linux protects applications and the kernel, how secure boot establishes a trusted boot chain, and how modern security technologies such as SELinux, TPM and kernel hardening are applied in production systems.

| Phase          | Coverage                                             |
|----------------|------------------------------------------------------|
| Phase 1        | Core Linux Kernel Fundamentals                       |
| Phase 2        | Memory & Process Management                          |
| Phase 3        | Filesystems & Storage                                |
| Phase 4        | Linux Networking                                     |
| Phase 5        | Interrupts, Timers & Deferred Work                   |
| Phase 6        | Advanced Device Drivers & Hardware Interfaces        |
| Phase 7        | Linux Security & Secure Boot                         |
| Estimated Size | 200–300 pages when expanded                          |
| Target         | Senior Embedded Linux / Kernel / Security Interviews |